

LIVE NOW

# Excelakuten

Automated repair for broken Excel files.

You don't need to know what's wrong — it finds out for you.

- 7-stage pipeline: load → diagnose → structure → clean → formula repair → validate → export
- You get a NEW file back — your original is never touched
- Fixes what it can — honestly reports what it can't

Python · FastAPI · openpyxl · LibreOffice · Azure Container Apps

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## THE PROBLEM

# Excel files nobody dares to touch

- #REF! errors and formulas pointing at deleted sheets and dead network drives
- Circular references that make Excel refuse to calculate
- Hidden sheets and columns that visible formulas silently pull numbers from
- Hardcoded values pasted straight over live formulas
- Text dates, numbers stored as text, stray whitespace — looks like data, isn't
- Corrupt XML: the file won't even open

*My test file: 106k+ formulas, 28k volatile, three sheets Excel itself wants to "repair".*

# Seven stages — each one logged

1

## Load

parses even corrupt files — CSV, .xls, .ods

2

## Diagnose

error map: hidden sheets, external links, broken chains

3

## Structure

finds tables, data types, data sources

4

## Clean

text dates → dates, "1 234.56" → numbers, dupes out

5

## Repair

#REF!, circular refs, pattern-based fill

6

## Validate

LibreOffice recalcs everything — new errors = stop

7

## Export

clean .xlsx + Report, Review and Changelog tabs

THE DEMO

# Drag in a file — watch the stages live

Reparera filen

3 s

Ren med varningar

Export\_avstamning\_KORRUPT.xlsx · 44 kB → 47 kB · klart på 3,0 s

✓

✓

✓

i

i

i

Ladda ner reparerad fil

Detta är en ny fil — din originalfil är orörd.

► Visa teknisk rapport

Ren med varningar

Kvarstående problem

X

X

X

X

X

Every stage reports in real time via the job API — no magic spinner.  
Files are processed in memory and never stored.

THE RESULT

98 seconds — everything accounted for

|    |                                                                                                                                                                               |                       |                                  |                              |        |     |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|----------------------------------|------------------------------|--------|-----|
| 1  | ÄNDRINGSLOGG                                                                                                                                                                  | Excel Repair Pipeline |                                  |                              |        |     |
| 2  | Ändringar av typen 'gissning'/'okänd' listas per cell nedan — de bör kontrolleras av en människa. Mekaniska ändringar ('säkra', t.ex. text→tal) summeras per blad längst ner. |                       |                                  |                              |        |     |
| 3  |                                                                                                                                                                               |                       |                                  |                              |        |     |
| 4  | ÄNDRINGAR ATT GRANSKA                                                                                                                                                         |                       |                                  |                              |        |     |
| 5  | Blad                                                                                                                                                                          | Cell                  | Före                             | Efter                        | Åtgärd | Typ |
| 6  | Inga ändringar kräver manuell granskning.                                                                                                                                     |                       |                                  |                              |        |     |
| 7  |                                                                                                                                                                               |                       |                                  |                              |        |     |
| 8  | SAMMANFATTNING — säkra ändringar                                                                                                                                              |                       |                                  |                              |        |     |
| 9  | Blad                                                                                                                                                                          | Antal                 | Åtgärd                           | Exempelceller                |        |     |
| 10 | Kvitton                                                                                                                                                                       | 19                    | cell rensad                      | A835, B835, C835, D835, E835 |        |     |
| 11 | Kvitton                                                                                                                                                                       | 300                   | text konverterad till datum      | B3, B7, B16, B18, B23        |        |     |
| 12 | Kvitton                                                                                                                                                                       | 19                    | överflödiga mellanslag borttagna | D30, D113, D125, D130, D208  |        |     |
| 13 | Mail från Anna                                                                                                                                                                | 1                     | överflödiga mellanslag borttagna | A9                           |        |     |
| 14 |                                                                                                                                                                               |                       |                                  |                              |        |     |
| 15 |                                                                                                                                                                               |                       |                                  |                              |        |     |

10,708

text values → numbers

2,447

text dates → real dates

4,105

circular references broken

40,134

formulas syntax-fixed

102,132

formulas recalculated — 0 new errors

56,651 safe changes summarised per sheet —  
uncertain ones listed per cell.

DEEPER THAN SYNTAX

# Correct, not just valid

*A formula can be syntactically perfect — and still return the wrong number.*

|       |
|-------|
| 1 240 |
| 860   |
| 5 112 |
| ...   |
| 430   |

1 500

**=SUM(B2:B41)**

=SUM(B2:B41) looks like it covers everything —

but row 42 was appended later.

| GRANSKA DESSA — Excel Repair Pipeline                                                                                                    |         |               |                                            |       |
|------------------------------------------------------------------------------------------------------------------------------------------|---------|---------------|--------------------------------------------|-------|
| Problem som hittats men inte ändrats — siffrorna här kan vara missvisande. Kolumnen Antal visar hur många liknande ställen som hittades. |         |               |                                            |       |
| Typ                                                                                                                                      | Blad    | Plats         | Detalj                                     | Antal |
| Summering missar data utanför området                                                                                                    | Kvitton | E903          | område E2:E886, data i: E887               | 1     |
| Summering missar data utanför området                                                                                                    | Kvitton | E904          | område E2:E100, data i: E101               | 1     |
| Text kvar i typad kolumn                                                                                                                 | Kvitton | kolumn Belopp | 59 celler ser ut som tal/datum men är text | 59    |

- Sums and COUNTAs whose range probably misses data appended outside it
- Formulas silently pulling numbers from hidden sheets
- Cells where formulas were overwritten with values
- External links to files that don't come along

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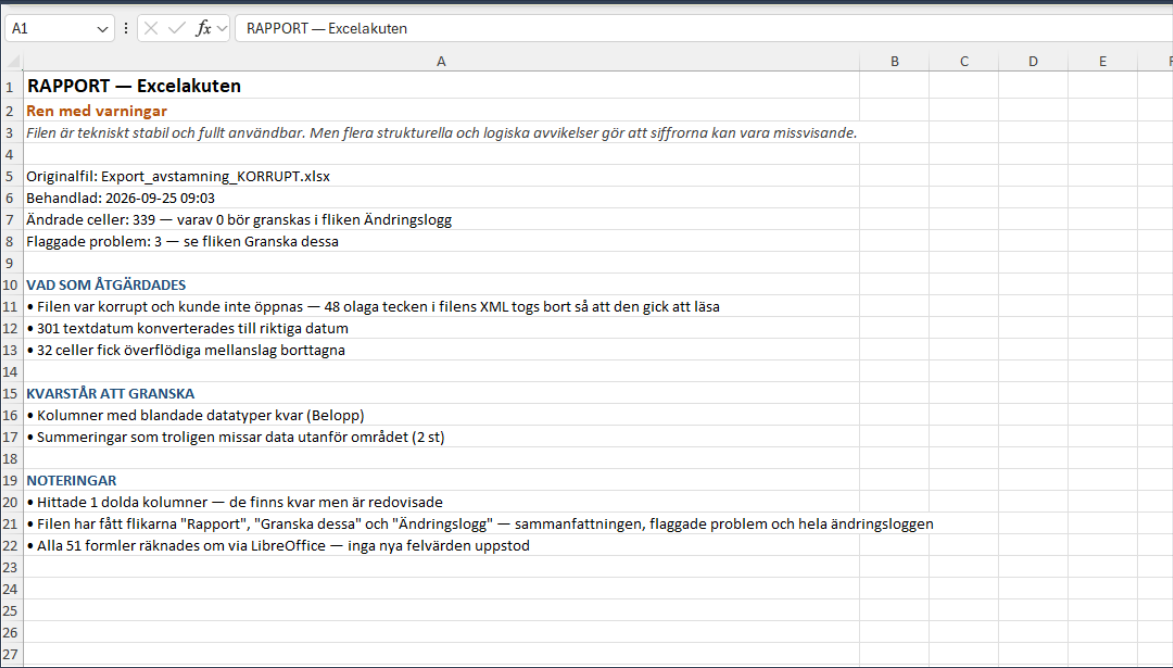
## SAFETY RULES

# What it refuses to touch

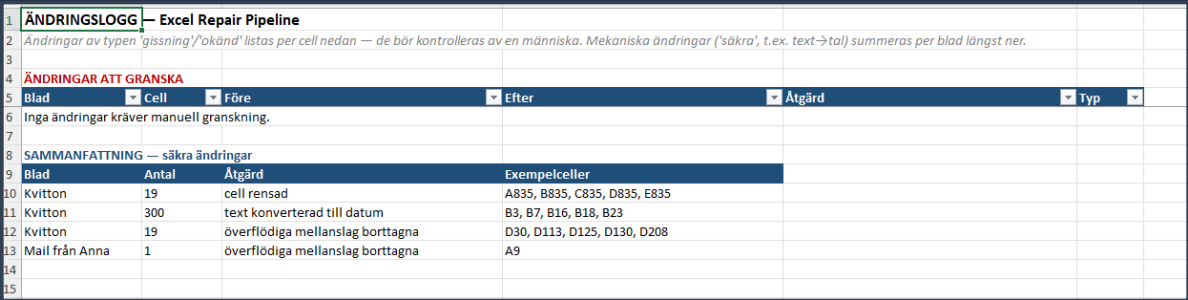
- ID columns like "00123" stay text — no silent conversion
- Rows and columns are never deleted if it would shift formulas — the cell is cleared and logged instead
- Table headers and table ranges are left untouched
- Hidden and protected sheets are never modified — they get flagged
- Reconstructed formulas always state the pattern they were built from

WHY I BUILT IT

It tells you what it CAN'T fix



The Rapport tab: the verdict, first in the file



The changelog: every cell, before/after

- Nothing is deleted silently — cleared cells are logged by reference
- An integrity check diffs every cell against the original — unexplained changes get flagged

Try it live:

[excelakuten.victoriousmoss-9ca9eab5.swedencentral.azurecontainerapps.io](https://excelakuten.victoriousmoss-9ca9eab5.swedencentral.azurecontainerapps.io)